



बड़ेबा RACING

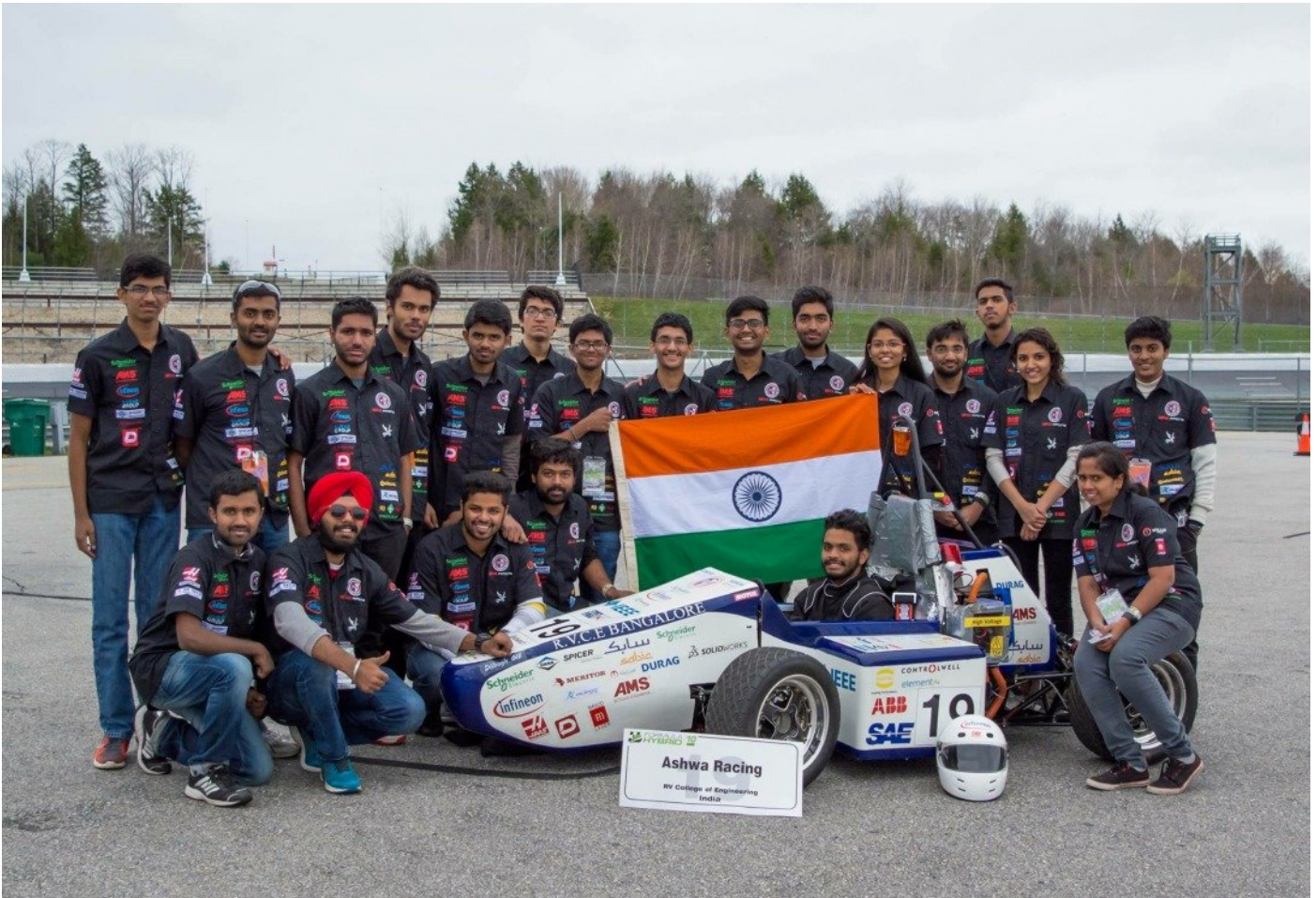
R.V. COLLEGE OF ENGINEERING

NEWSLETTER - JULY 2016





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Dear readers,

The 2016 race-season has been a year full of positives for the team and we are pleased to inform you that we have bagged 2nd position in Design and Project management events, thus finishing 4th overall, among 28 teams that participated at Formula Hybrid '16. The event was held at the New Hampshire Motor Speedway from 2nd to 5th May, which is one of the most scintillating tracks in the NASCAR racing history.

Meanwhile, it's the last month of testing our combustion prototype, RZX6-C, after which it'll be shipped to Czech Republic. We've also started the designs of our 2017 prototypes and are looking forward to a summer of manufacturing.



The Formula Hybrid competition is an interdisciplinary design and engineering challenge for undergraduate and graduate university students. They must build an electric or plug-in hybrid race car and compete in a series of events. Teams from USA, Canada, Turkey, Sweden, Russia, Taiwan, Spain and India have participated since the inception of the event.

The Formula Hybrid event was founded and is run by the Thayer School of Engineering at Dartmouth since 2006. Formula Hybrid takes place each spring at the New Hampshire motor speedway in Loudon, NH. The competition is part of the Society of Automotive Engineers Collegiate Design Series and is regarded as the most complex and dynamic of the series.

The Formula Hybrid event has been a good challenge to the team, forcing us to look into the 'green' way of racing. Once we master the hybrid, we look to diversify into the electrical category as well.



RZX6-H



We competed at Formula Hybrid 2016 with the elegant RZX6-H. RZX6-H runs on a series hybrid architecture. It is powered by an Agni-95R motor which drives the wheels. An Agni-95 motor, coupled with a Pulsar 220cc bike engine is used as a generator to charge the batteries. The prototype which weighs 318kg features a space framed chassis and uses Lithium ion battery with custom BMS (Battery Management System). It has a torsen differential, an ingenious rear actuated pedal assembly that reduced weight and overhang, unequal A-arms with ZF dampers and was built to comply with international safety standards.

The main design goal while building this race-car was performance and safety. The event challenges us to abide by the 35.5 MJ energy limit rule and suitably decide on the architecture to increase performance.

The prototype which was designed, manufactured and assembled in R.V. College of Engineering in 10 months and shipped to US on the 10th of March reached the event site on 28th of April.



Team at NYC



After a flight that stopped in 4 countries, we finally reached New York on the 26th of April, after which we met Sam and Patrick from TED and briefed them about our project. They appreciated our work and gave us a book on Ted fellow's adventure.



We also visited the Indian Embassy and thanked the Consulate General Mrs. Riva Ganguly Das for helping us with all problems faced in the US, including shipping of the car.



Event Preparations



We proceeded towards Plymouth, New Hampshire via Boston, by bus. This time, we had taken measures in order to clear the US customs before hand so that we get sufficient time to work on the car before going to the event.

We worked on the car for three days and made the car event ready. Two of our notable Alumni, Shubham Saurav and Vikram Raghavan joined us at the event and boosted our morale by being with us through thick and thin and guiding us throughout the event.



Static Events

Project Management presentation



Here every team is evaluated on how effectively and efficiently the team managed the project with the given deadline and available resources. We secured 2nd position in Project Management presentation with a score of 94.17.



Design presentation



In this event, designs are evaluated based on performance, innovation and conceptual knowledge of the team. After the initial round of evaluation, we were selected for the finals where we presented our design to two different panels of judges and once again, we bagged the 2nd position by scoring 164 out of 200 points.



Technical Scrutiny



Before entering the dynamic events, the car is extensively scrutinized to ensure if it is safe to go on track. This includes three phases:

- Pre technical inspection- Electrical
- Technical inspection- Mechanical
- Final Technical inspection- Electrical

With a few minor changes, we cleared the Pre-technical inspection on the second day of the event.

With a good knowledge of mechanical scrutiny due to our decade long experience in the combustion category, we cleared mechanical scrutiny with relative ease on the third day.

We were left with the hardest scrutiny, which was Final Tech Inspection. At around noon on Day 4, we cleared the final hurdle.



Visitors at the event site



Mr. Dilbagh Gill, Team Principal of Mahindra Racing Formula E-team, paid us a visit at the event site and gave invaluable inputs about the ways to manage a race team and its structure. The team thanked him for his time and views by giving him a memento. He gifted us Mahindra Racing goodies.



The Indian Embassy had also introduced us to **Mr. Rajat Bhatnagar**, Advisor of technical Solutions Inc. who created a very conducive environment which made us feel like home in the foreign nation. He got along the dynamic, car fanatic, President of the company, **Mr. Chris Harris** who encouraged us and cheered for us by staying throughout the design finals.



Pragati aunty has been the team's pillar of support during the event. Starting from helping out with accommodation and food to ensuring good health of the team members, she took care of us like a mother. We are grateful to have her as our ashwa friend and have built a long lasting relationship with her. The memories shared with her will be cherished forever. Once again thank you for all the love and care.



We are thankful to Mr. Michael Drew, owner of Frontenac lodge, for our accomodation at New Hampshire. It was because of him that we got a place to work on the car. He also helped us in our travel from the lodge to the event site. Be it any time in the day as well as in the night, he helped us by driving us around in his car. He even came down to the event site one day and stayed till our scrutiny was completed in the midnight. We are grateful to him for helping us and hope to continue this relationship with him.



Mr. Michael Bouchard is a military scientist from Massachusetts Institute of Technology. Sir Michael Bouchard, fondly called as Mike, owner of the Tenny Ski Mountain, helped us immensely by providing us his military truck to transport the cargo box from the event site to the garage on the Tenny Ski Mountain. He also allowed us to use his garage to assemble the car from its bone dry state to the running state. Had we not worked in the garage, it wouldn't have been possible for the team to utilize the added day in the Pre-Electrical Scrutiny, which was one of the most important factor in clearing the Electrical Scrutiny.

Apart from this, he also helped Mr Michael Drew to arrange a bus for us to travel from the event site to the lodge.

He also invited us to visit his mountain and took us around the mountain. We thank him for all the help he provided and wish to meet him every time we travel to the USA.



Victory!



After all the challenges, we secured 4th position overall. The team bettered the previous results by climbing the position ladder by three steps. With an unforgettable event experience and with a lot of suggestions and guidelines to be passed down to the juniors, the team returned to its motherland, victorious!

The team is happy with its performance at this year's Formula Hybrid event. This will encourage more teams from India to be part of such events, thus helping showcase Indian Talent on an international platform. Our notable highs proved the ever increasing competitiveness of Indian teams at the competition. We would like to thank all our sponsors, our alumni, RV College of Engineering, Rashtriya Shikshana Samithi Trust (RSST), for supporting us. Also, without the help of The Consulate General, **Mrs. Riva Ganguly Das**, **Ms. Uma Anand**, **Mr. Rajat Bhatnagar**, **Mr. Chris Harris**, **Mr. Micheal Drew**, **Mr. Micheal Bouchard**, **Ms. Pragati Narahari**, **Mr. Dilbagh Gill**, Our faculty Advisor **Dr. Ravindra S. Kulkarni**, our Principal **Dr. K.N. Subramanya**, our Steering Committee members **Arunraj Subbaraj** and **Sidharth Mandal** and our active Alumni **Shubham Sourav**, **Vikram Raghvan**, **Nikhil Jali** and **Rajit Malik**, our event wouldn't have been successful.



Improvements in Ergonomics



A significant improvement has been made in the ergonomics setup over that of the previous years'. The older setup was limited to variables like pedal location, front hoop height and main hoop height; while the new setup allows us to change additional parameters such as cockpit inclination, main hoop inclination, steering wheel inclination, shoulder harness bar height and seat inclination.

The cockpit is the interface between the car and the driver and must be ergonomically designed to allow the driver to maneuver through all the course obstacles.

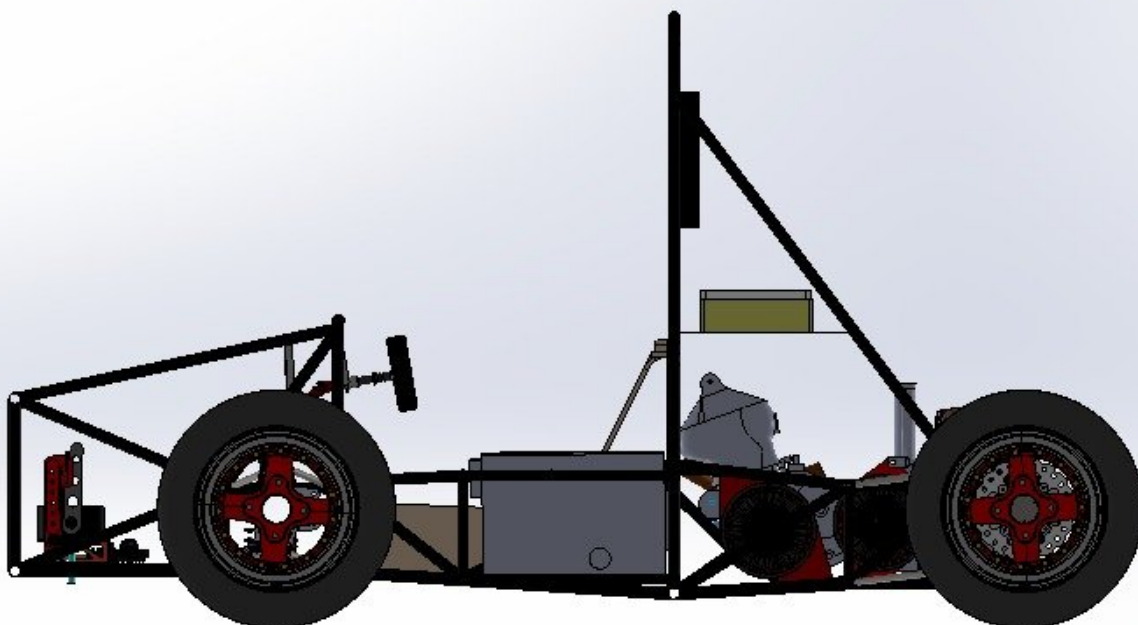
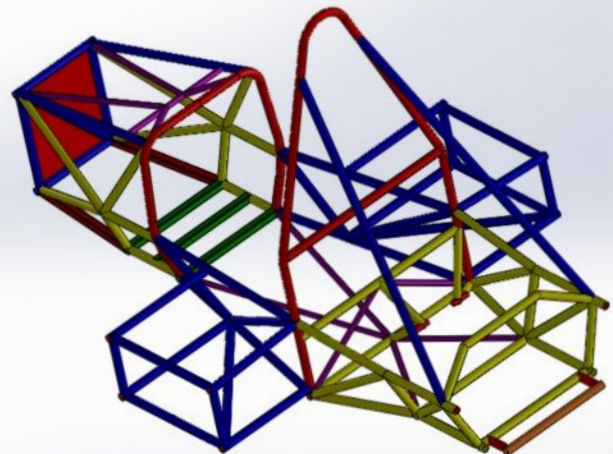
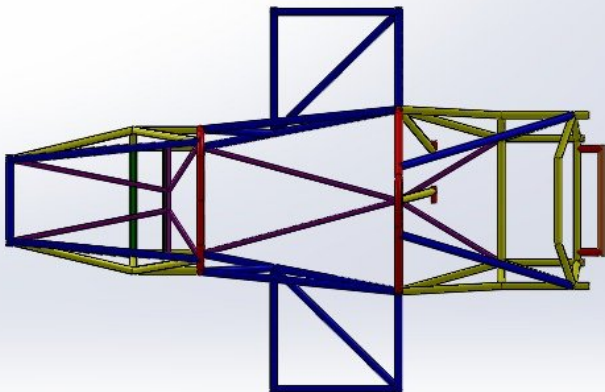
The new setup enables us to decide the inclination of the main hoop for better packaging and also helps us improve driver visibility.



Design phase of RZX7-H

The design phase for the 2017 race season began in the first week of April. To keep up with deadlines, the design phase of the hybrid prototype was shortened from 3 months to 1 month so that we get more time for testing the car.

Since Ashwa Racing believes in evolution over revolution, 5% increment in design changes were brought about. We have switched from simple gear train to compound gear train, to allow a higher reduction ratio and hence lower acceleration times.





Procurement of Steel tubes



Steel Mart provided us with steel tubes this year to build our chassis. They were generous to provide us a discount of 50% and we're grateful to them.

We are thankful to them for providing the steel tubes at a discounted price and delivering them in a short time too.



Aerodynamics



The team has stepped into a new tributary of building race cars. Realizing the importance of aerodynamics and its performance benefits, research is being carried out on the same.

Prototypes of front and rear wings which were designed to generate additional 30kgf of downforce, were fabricated in-house and assembled on the car. Testing has just begun and we are conducting preliminary tests before heading to the wind tunnel.



Alumni meet 2016



We displayed the combustion prototype at the alumni meet, at our college. It was a pleasant experience to interact with former members of the college and receive inputs on the prototype. A huge thanks to the alumni for their presence as well as their insight.





Werner Finley Annual Day



Our hybrid partner, Werner Finley had their official annual day on 7th of April. As a token of gratitude for their unending support towards our endeavour, we displayed our latest prototype, the RZX6-C at their fun-filled event.

We thank Werner Finley for giving us the opportunity to showcase our prototype. We had an amazing time there. Cheers!





Alumni focus



We would also thank our 2008 team member Vikram Raghavan, who is currently working at Robert Bosch GmbH, for working closely with the electrical team to debug various electrical issues with the car. Although being from the mechanical background, his management abilities to debug issues and solve problems in a structured fashion directed the team to identify and iron out major unknown electrical complications. With exemplary people skills, he prevented the rise of tensions at the workplace and kept the team jovial. Even working sleeplessly overnight did not seem stressful with Vikram around. It was a great experience to work with Vikram.

Our 2009 team member Shubham Saurav, who is currently pursuing his Master's degree at University of Michigan, helped our team at the Formula Hybrid '16 event. Shubham, with his profound interest in vehicle dynamics and innate love for Ashwa Racing, strived with the team to prepare the car at the event. He directed the team and managed workforce which increased the productivity of the team. With his familiarity with the rules and technical inspection, he led the team to clear scrutiny with ease. Shubham also boosted the team's morale, ensured good health of the team members at the event, and kept everyone at the peak of their industry.

The team looked up at and learnt a lot from our steering committee member, without whom, we would not have succeeded as we have.





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Guhring India is a 100% owned subsidiary of Guhring KG. It is a leading manufacturer of high quality, rotaprecision cutting tools. Main business activities are trading of standard tools, manufacture of custom made tools and related services.

The team is thankful to Guhring India for their support.

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The team is extremely thankful to Sansera engineering for their continued support. This partnership, now older than a decade is of great value to the team.

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DURAG

Durag Group has been operating successfully in India for over 30 years. Some of the services provided by Durag India are Local manufacturing, Assembly & testing, Advisory Services and sales, Commission and Installations, Maintenance and servicing. Durag Group became a part of the Ashwa Racing Family in this year and we are hoping for a long lasting relationship with the company in the coming times.

We would like to thank Felsomat for their support and belief they have shown at the time we needed the most. Felsomat develops innovative machine tool technology for Economic production worldwide.



Mouser Electronics, a subsidiary of TTI, Inc., is part of Warren Buffett's Berkshire Hathaway family of companies. Mouser is an award-winning, authorized semiconductor and electronic component distributor, focused on the rapid introduction of new products and technologies to electronic design engineers and buyers. Mouser.com features more than 4 million products online from more than 600 manufacturers. Mouser publishes multiple catalogs per year providing designers with up-to-date data on the components now available for the next generation of electronic devices. Mouser ships globally to over 500,000 customers in 170 countries. The team is indebted towards Mouser Electronics for their assistance and support.

TE Connectivity (TE), we develop connectivity and sensor solutions that enable automation and connectivity in the home. Their solutions fit into the tiniest spaces and stand up to rigorous daily use.

We thank them for the financial as well as the component support that they have given us.





KUMMAYA DESIGN STUDIOS

Kummaya is a design firm based out of Bangalore India. Kummaya specialize in Industrial design, Mechanical engineering design and product development. Their areas of expertise spans from electronic products to medical devices. We would like to take this opportunity to thank Kummaya for their support.



SureWaves, through its ground-breaking technology, is mapping, aggregating and consolidating diverse and dispersed consumption markets and consumers across India, through information, technology and knowledge tools, to usher an era, where markets, media and consumption converge to deliver unprecedented growth through inclusion and innovation.

A heartfelt thank you for the sponsorship.

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They provided us with dial gauges which were used while performing the torsional test to measure the stiffness of the chassis



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